

Low Poly City FREE

Modular city kit - documentation and setup guide

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1. What this pack is

50 low-poly city models, built to snap together on one 4.00 m grid: modular building facades; pitched and flat roofs; doors and windows; roads, junctions and kerbs; pavements and ground; street furniture; vehicles; trees and planting; shopfronts and signage; civic buildings; industrial buildings; utilities; fences and barriers; interior depth behind glazing; and the dressing that makes a street look lived in.

Everything is authored in metres at real-world scale, so a model dropped into a scene is the size the thing would be. A car is 4.40 m long and a storey is 3.00 m, which is what they are.

Models	50
Triangles	24 to 756, averaging 254
Materials	3 - opaque, glass, emissive
Textures	1024x1024 colour atlas, plus a metallic/smoothness atlas in the same layout, and masked glass and emissive maps
Format	FBX

2. Importing

Import the package. Everything lands under Assets/LowPolyCityFree.

The models arrive already textured: each FBX names the pack's materials, and its importer settings bind them by name.

If the project uses URP or HDRP, the pack converts its materials on first import. If anything looks magenta, run the convert menu item below.

Then run **Tools > Low Poly City > Build Prefabs** once.

3. The grid

The kit's one promise is that the pieces meet. Every join in the demo scenes is measured tight to within a tenth of a millimetre.

Everything drops at $y = 0$ - walls, roofs, roads alike. A wall's bottom 0.20 m is buried in the floor slab, which is what lets one placement height work for every piece and gives vertical tolerance on uneven ground.

Tile	4.00 x 4.00 m in plan
Storey	3.00 m floor to floor
Wall piece	4.00 m long, 0.30 m thick, 3.00 m tall
Clear height	2.80 m, floor surface to ceiling
Domestic door	0.90 x 2.10 m
Commercial door	1.80 x 2.40 m
Garage door	2.40 x 2.20 m
Window	1.20 x 1.40 m, sill 0.90 m above the floor

Walls are dressed on BOTH faces - render outside, plaster inside - so one piece serves as an external wall and as the party wall of a two-room plan, rather than two pieces back to back.

4. Roofs

Every pitched roof in the pack is a 1:2 slope - a panel covers 2.00 m of plan and rises 1.00 m, which is 26.6 degrees.

One pitch and no other is what makes hips and valleys tile. Run and rise are both exact numbers, so a ridge height is always a whole multiple and a hip mitres at 45 degrees in plan.

Flat roofs are the other half of a city and use a parapet rather than a pitch.

Slope panel	4.00 m wide, 2.2361 m along the slope
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Eaves overhang	0.40 m beyond the wall face
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Flat roof parapet	0.60 m
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5. The street

Road pieces are derived from the same module as the buildings, so a junction never needs a part tile.

A road tile placed at $y = 0$ has its driving surface at 0.20 m, the same height as a floor slab's walkable face, so a vehicle and a character stand on the same datum.

Carriageway	8.00 m kerb to kerb, two 3.50 m lanes
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Junction	8.00 x 8.00 m
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Pavement	2.00 m wide, standing 0.15 m proud of the road
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Street corridor	12.60 m, building face to building face
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Lane markings are separate pieces laid on the road rather than cut into it, so a junction can be marked any way a project needs.

6. Pivots and orientation

Every model is pivoted at the bottom-centre of its bounds, so dropping it at $y = 0$ stands it on the ground.

Doors, gates and shutters are pivoted on their HINGE EDGE instead, so rotating the prefab swings the leaf on its hinges rather than spinning it about its middle.

Every model faces Unity +Z with no rotation on its root object, so `transform.forward` points out of the front of a building or a vehicle.

7. Materials and recolouring

The pack is three materials and one real texture. Every face is UV-mapped to the centre of one cell in the colour atlas, and the metallic/smoothness atlas uses the same cell layout, so steel reads as metal and render does not.

Recolour the whole kit by editing `LPC_Palette.png`. Every model that used a swatch changes with it - repaint a street by editing one row of one PNG. Keep point filtering on: a face sits at a cell CENTRE, and bilinear filtering across a cell edge would blend two colours along every seam.

Glazing is a genuinely transparent material, and every glazed model carries an opaque panel behind it so a window reads as a window rather than as a hole. If you replace the glass material with an opaque one,

that backing is what you will see.

The emissive material is for lit signs, lamp heads and traffic lenses. Lit and unlit versions ship as separate models where it matters, so a street can be switched from day to night by swapping prefabs.

8. Editor menu reference

All of these live under **Tools > Low Poly City**.

8.1 Build Prefabs

Walks the Models folder and writes one prefab per FBX, each carrying a mesh renderer, a mesh collider and the pack's shared materials. Run it once after import; it is safe to run again.

8.2 Build Library Scene

Lays every model out on a grid in a new scene, so the whole kit can be looked through without dragging prefabs one at a time.

8.3 Convert Materials to This Pipeline

Detects the active render pipeline and converts the pack's three materials to it. It runs by itself on first import and decides per material by reading that material's shader, so importing into a second URP or HDRP project converts correctly too.

9. Render pipelines

Built-in, URP and HDRP are supported. The pack ships on the Built-in Standard shader and converts on import.

The metallic/smoothness atlas is carried across under each pipeline's own property names: `_MetallicGlossMap` on Built-in and URP, and `_MaskMap` on HDRP, where its green channel carries ambient occlusion.

10. Performance

Models run 24 to 756 triangles, averaging 254. The modular walls, roofs and road tiles, which repeat most, are deliberately the cheapest things in the pack - a plain wall is under 150 triangles.

That is a smaller budget than a dungeon or interior kit uses, and it is deliberate: a room holds about forty modules and a city block holds four hundred, so the load is carried by the section of each piece rather than by added geometry.

Three materials means three draw calls for a fully dressed street before batching. Enable GPU instancing on the materials for large scenes: the kit is built from repeated pieces, which is the case instancing is for.

11. Support

If something does not behave as this guide says, reply on the Asset Store page for this pack with your Unity version and render pipeline.

The complete kit - every category here plus shopfronts, civic buildings, industrial, utilities and the rest of the vehicles - is on the same publisher page.